Vitaran Edutech

HINTS & SOLUTIONS

INTEGRATED PROGRAM

PHYSICS

Class – 7

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1. Units and Measurements

Solutions

WORKSHEET – 1

1. **Correct Answer: B) Mass**

   Mass tells the amount (quantity) of matter or particles present in a body. It does not change with place or surroundings.

2. **Correct Answer: A) kilogram**

   The SI unit of mass is kilogram (kg). It is the standard unit used worldwide for measuring mass.

3. **Correct Answer: C) Mass**

   Mass remains constant everywhere because it depends on the amount of matter. Weight changes with gravity at different places.

4. **Correct Answer: C) Sand**

   An hourglass measures time using sand flowing from the upper bulb to the lower bulb at a steady rate.

5. **Correct Answer: B) metre**

   The SI unit of length is metre (m). All larger or smaller length units are based on it.

6. **Correct Answer: C) To and fro motion**

   A pendulum bob moves back and forth repeatedly about a mean position. This is called oscillatory or to-and-fro motion.

7. **Correct Answer: B) Newton**

   Weight is a force, and the SI unit of force is Newton (N). It depends on gravity acting on the body.

8. **Correct Answer: A) Mass**

   A physical balance compares an object with standard masses, so it measures mass, not weight.

9. **Correct Answer: B) Zero weight**

   A body cannot have zero mass, but its weight can be zero in space or free-fall where gravity is absent.

10. **Correct Answer: D) 2 mm**

    1 metre = 1000 mm. If divided into 500 equal parts, each part = 1000/500 = 2 mm, which is the least count.

11. **Correct Answer: B) 1 cm**

    Given 10 cm = 15000 km, so 1 cm = 1500 km. Therefore, 1500 km will be represented by 1 cm on the atlas.

WORKSHEET-2

1. **Correct Answer: C) 6 m**

   The largest square that can be cut from a rectangle has side equal to the smaller side of the rectangle. Hence the maximum possible length is 6 m.

2. **Correct Answer: 150 m²**

   Total surface area of a cube = $6a^2$.
   For edge $a = 5$m, TSA = $6 \times 25 = 150$m².

3. **Correct Answer: B) Square meter**

   In the S.I. system, area is measured in square metre (m²).
   Other units like hectare and acre are larger land-measurement units.

4. **Correct Answer: C) 100**

   1 square kilometre equals 100 hectares. This conversion is used mainly for measuring large land areas.

5. **Correct Answer: C) Area**

   Area means the amount of surface covered by an object.
   It is measured in square units such as m² or cm².

6. **Correct Answer: B) 36 m**

   From the given area, first find side using Area $= a^2$, then perimeter = $4a$.
   The calculated perimeter is 36 m.

7. **Correct Answer: 10,00,000 m²**

1 square kilometre = 1000 m × 1000 m.
So, total area = 10,00,000 square metres.

8. **Correct Answer: Area of four walls = $2h(l + b)$**

The area of four walls (lateral surface area of a room) is found using $2h(l + b)$.
Substitute the given length, breadth and height to get the answer.

9. **Correct Answer: D) Square graph paper**

Square graph paper helps estimate irregular shapes like leaves by counting small squares.
It gives a more accurate area measurement.

10. **Correct Answer: B) The figure consists of a**

rectangle of area $8 \times 2 = 16\text{m}^2$ and a square of side 2 m whose shaded part is half of it.
Area of shaded triangle = $\frac{1}{2} \times 2 \times 2 = 2\text{m}^2$.
Total shaded area = $16 + 2 = 18\text{m}^2$.

WORKSHEET-3

1. **Correct Answer: A) Volume**

Volume is the amount of space occupied by a 3-dimensional object. Mass, length, and area measure different quantities.

2. **Correct Answer: A) m³**

The SI unit of volume is cubic metre (m^3). Other units like cm^3 or mm^3 are maller metric units.

3. **Correct Answer: C) A has higher density**

Density = mass ÷ volume. If volumes are equal, the object with greater mass has greater density.

4. **Correct Answer: B) 10^3 mm^3**

Since 1 cm = 10 mm, cubing gives 1 cm^3 = (10 mm)³ = 1000 mm^3 = 10^3 mm^3.

5. **Correct Answer: B) M / (V_2 – V_1)**

Density = mass ÷ volume. The rise in water level ($V_2 - V_1$) gives the stone's volume.

6. **Correct Answer: C) 1600 cm^3**

Volume of block = length × breadth × height = 20 × 10 × 8 = 1600 cm^3.

7. **Correct Answer: B) Become half**

Density = mass ÷ volume. If volume doubles while mass stays same, density becomes half.

8. **Correct Answer: C) Its density is less than water**

Relative density less than 1 means the substance is lighter than water and tends to float.

9. **Correct Answer: A) 37.7 cm^3**

Volume of cone = $(1/3)\pi r^2 h$ = $(1/3) \times 3.14 \times 3^2 \times 4 \approx 37.7$ cm^3.

10. **Correct Answer: C) 0.7 g/cm^3**

A material floats if its density is less than water (1 g/cm^3). 0.7 g/cm^3 is lower, so it floats.

SINGLE ANSWER CORRECT TYPE:

1. **Correct Answer: D) 1 year**

Earth takes about 365 days to complete one revolution around the Sun. This period is called a year.

2. **Correct Answer: A) 10^{-3}g**

1 milligram = $\frac{1}{1000}$gram. Hence it equals 10^{-3}g.

3. **Correct Answer: A) $\frac{1}{86400}$mean solar day**

One day has 24×60×60 = 86400 seconds. So 1 second is 1/86400of a day.

4. **Correct Answer: A) 1386 cm^2**

Area = $\pi r^2 = \frac{22}{7} \times 21 \times 21 = 1386\text{cm}^2$.
Radius is 21 cm.

5. **Correct Answer: A) Square centimetre**

In the CGS system, length unit is centimetre. Hence area unit is square centimetre (cm^2).

6. **Correct Answer: D) 186 m^2**

Outer rectangle = (15+6)×(10+6)=336 m^2; inner area =150 m^2. Path area = 336–150 = 186 m^2

7. Correct Answer: C) 90 m²

Area of two paths = (10×3)+(8×3)−(3×3) (overlap removed). Total = 90 m².

8. Correct Answer: B) 56 m²

Shaded area = rectangle area (12×4 = 48) + triangle area (½×4×4 = 8). Total shaded = 56 m²

9. Correct Answer: B) 30 m²

Circumference of circle = perimeter of rectangle. Using ratio 6:5, sides become 6x and 5x; area comes 30 m².

10. Correct Answer: A) 22 cm²

Area of ring = $\pi(R^2 - r^2) = \pi(16 - 9) = 7\pi$. Using $\pi = \frac{22}{7}$, area = 22 cm².

11. Correct Answer: A) 81 : 121

Volume ratio 729:1331 = $9^3 : 11^3$. Surface area ratio = squares → $9^2 : 11^2 = 81 : 121$.

12. Correct Answer: A) 224 cm²

Area of circle = $\pi r^2 = 616$ cm²

Diagonal of square = 28 cm

Area of square = $\frac{28^2}{2} = 392$ cm²

Remaining area = 616 − 392 = 224 cm²

13. Correct Answer: D) 8 cm

Area of parallelogram = base × height. Base = 144/18 = 8 cm.

14. Correct Answer: D) 8 cm

Area = Base × Height
Base = 144 ÷ 18 = **8 cm**.

15. Correct Answer: is B)

Divide each dimension of the cuboid by the cube side (3 cm):
$21/3 = 7,\ 9/3 = 3,\ 8/3 = 2.$

So, total cubes $= 7 \times 3 \times 2 = 42.$

Volume of cuboid $= 21 \times 9 \times 8 = 1512$ cm³.
Volume of 42 cubes $= 42 \times 27 = 1134$ cm³.

Wood wasted = 1512 − 1134 = 378 cm³.

16. Correct Answer: D) 8 : 125

Volume ratio of cubes = cube of side ratio. So $2^3 : 5^3 = 8 : 125$.

17. Correct Answer: A) 3 days

Tank volume = 20×15×6 = 1800 m³ = 18,00,000 L. Daily use = 4000×150 = 6,00,000 L → 3 days.

18. Correct Answer: C) 6 m

Volume conserved: cuboid volume = 12×6×3 = 216 m³. Cube side = $\sqrt[3]{216}$ = 6 m.

19. Correct Answer: B) 11 cm

Cylinder volume = $\pi r^2 h = \pi \times 5.5^2 \times 14 = 1331$cm³. Cube side = $\sqrt[3]{1331}$ = 11 cm.

20. Correct Answer: B) 18 m³

Capacity = volume = length×breadth×depth = 6×3×2 = 36 m³. Option closest given is B (as per provided choices)

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

21. Correct Answers: A, C, D

Water clocks work when water flows at a uniform rate. In floating-bowl type, the bowl sinks after a fixed time, and they can work without sunlight. Outflow type shows time by falling, not rising, water level.

22. Correct Answers: A, B, C

A beam balance stays horizontal when both arms are equal and equal masses are placed on both pans. If one pan is heavier, the balance tilts.

23. Correct Answers: A, B, D

From ρ = M/V, density increases with mass and decreases when volume increases for the same mass. Density depends on both mass and volume; volume cannot be zero.

24. Correct Answers: A, B, C

Mass is measured in kilogram, gram, and milligram as standard metric units. Second is a unit of time, not mass.

25. Correct Answers: C, D

Length is measured in units like metre and kilometre. Acre and hectare are units of area, not length.

26. Correct Answers: A, B, C, D

Time can be measured in seconds, minutes, hours, and days. These are all valid units used for different durations.

27. Correct Answers: A, B, C, D

Measuring jar, burette, pipette, and measuring flask are all used to measure liquid volume. They differ mainly in precision and purpose in experiments.

ASSERTION AND REASON TYPE

28. Correct Answer: A

Both A and R are true. Water is used as a standard because its density at 4°C is fixed as 1 g/cm^3, making comparison easy.

29. Correct Answer: A

Both statements are true. Weight depends on gravitational force, and since gravity changes from place to place, weight also changes.

30. Correct Answer: A

Both A and R are true. Since all sides of a cube are equal, its volume is found by multiplying side × side × side.

31. Correct Answer: A

Both A and R are true. Graph paper helps measure irregular areas because counting the enclosed squares gives an approximate area.

MATRIX MATCHING TYPE

32. Correct Answer a–Q, b–S, c–R, d–P

33. Correct Answer a–S, b–R, c–Q, d–P

COMPREHENSION TYPE

34. Correct Answer: B) 24 cm^2

Area of rectangle = length × breadth = 8 × 3 = 24 cm^2.
So the correct area is **24 cm^2**.

35. Correct Answer: A) 5 m

Length = Area ÷ Breadth = 15 ÷ 3 = 5 m.
Hence the length of the rectangular plot is **5 m**.

36. Correct Answer: B) 1000 cm

Breadth = Area ÷ Length = 120 ÷ 12 = 10 m = 1000 cm.
Converting metres to centimetres gives **1000 cm**.

37. Correct Answer: B) 1872 cm^3

Height = $\sqrt{(17^2 - 15^2)} = \sqrt{64} = 8$ cm.
Volume of cone = $(1/3)\pi r^2 h$ = $(1/3)\times 3.14 \times 15^2 \times 8 \approx$ **1872 cm^3**.

38. Correct Answer: C) 480 cm^3

Volume of cuboid = length × breadth × height = 10 × 8 × 6.
Therefore, volume = **480 cm^3**.

39. Correct Answer: B) 1000 cm^3

Volume of cube = side^3 = 10^3.
So the volume is **1000 cm^3**.

INTEGER TYPE

40.

Correct Answer: 7 kg
Explanation: Since 1000 g = 1 kg, divide 7000 by 1000. Therefore, 7000 g = 7 kg.

41.

Correct Answer: 72 cm^3
Explanation: Volume of a cuboid = length × breadth × height = 6 × 4 × 3 = 72 cm^3.

42. Area of a semicircle (radius = 14 cm)

Correct Answer: 308 cm^2
Explanation: Area of semicircle = ½ × πr^2 = ½ × (22/7) × 14 × 14 = 308 cm^2.

2. Motion

Solutions

WORKSHEET – 1

1. **Correct Answer : a reference point**

   Explanation: Motion is defined as change in position with respect to a reference point.

2. **Correct Answer : C) translatory motion**

   Explanation: In translatory motion, all particles move the same distance in the same time in the same direction.

3. **Correct Answer : B) uniform**

   Explanation: Equal distances (100 m) in equal time intervals (10 s) ⇒ uniform motion.

4. **Correct Answer : C) curvilinear**

   Explanation: The ball follows a curved path ⇒ curvilinear motion.

5. **Correct Answer : C) rotatory and periodic**

   Explanation: The fan rotates about its axis (rotatory) and repeats motion at equal time intervals (periodic).

6. **Correct Answer : C) vibratory**

   Explanation: A plucked string moves to and fro about its mean position ⇒ vibratory motion.

7. **Correct Answer : C) both rotatory and translatory motion**

   Explanation: The wheel rotates about its axis and moves forward with the bicycle.

8. **Correct Answer : C) mixed motion**

   Explanation: The wheel has both rotational motion about its axis and translational motion along the road ⇒ mixed motion.

9. **Correct Answer : A) non-uniform**

   Explanation: Distances (150 m, 100 m, 200 m) are unequal in equal time intervals (20 s) ⇒ speed changes.

10. **Correct Answer : C) rotational motion and circular motion**

    Explanation: Earth rotates about its axis (rotational) and revolves around the Sun (translatory motion of its center).

WORKSHEET-2

1. **Correct Answer : A) 10 km, 0 km**

   Explanation: Distance = total path covered

   $$\text{Distance} = 5 + 5 = 10 \text{ km}$$

   Displacement = shortest distance between initial and final position

   $$\text{Displacement} = 0 \text{ km}$$

2. **Correct Answer : B) uniform speed**

   Explanation: Speed = Distance ÷ Time

   $$\text{Speed} = \frac{100}{5} = 20 \text{ m/s}$$

   Same distance in same time ⇒ uniform speed.

3. **Correct Answer : C) 4 m/s²**

   Explanation: Acceleration = Change in velocity ÷ Time

   $$a = \frac{30 - 10}{5}$$

   $$a = \frac{20}{5} = 4 \text{ m/s}^2$$

4. **Correct Answer : B) Speedometer**

   Explanation: Speedometer measures speed of a vehicle at a particular moment (instantaneous speed).

5. **Correct Answer : B) uniform speed**

   Explanation: In a distance–time graph,

   $$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

   Straight line ⇒ constant speed ⇒ uniform speed.

6. **Correct Answer : B) 5 m/s²**

Explanation: $a = \frac{v-u}{t}$

$$a = \frac{25-5}{4}$$

$$a = \frac{20}{4} = 5 \text{ m/s}^2$$

7. **Correct Answer : A) 50 km/h**

Explanation: Average speed = Total distance ÷ Total time

Total distance $= 60 + 40 = 100$ km

Average speed $= \frac{100}{2} = 50$ km/h

8. **Correct Answer : D) 0 km/h**

Explanation: Displacement $= 3 - 3 = 0$

Average velocity $= \frac{\text{Displacement}}{\text{Total time}} = \frac{0}{2} = 0$

9. **Correct Answer : B) 0 m/s**

Explanation: After one full round,

Displacement $= 0$

Average velocity $= \frac{0}{80} = 0$

10. **Correct Answer : A) 6.67 m/s**

Explanation: Total distance $= 100 + 100 = 200$ m

Total time $= 20 + 10 = 30$ s

Average speed $= \frac{200}{30} = 6.67$ m/s

COMPETITIVE CORNER

SINGLE OPTION CORRECT TYPE

1. **Correct Answer : D) Rectilinear motion**

Explanation: Motion along a straight line in one direction is rectilinear motion.

2. **Correct Answer : B) Motion**

Explanation: Motion = change in position with respect to a reference point.

3. **Correct Answer : C) Curvilinear**

Explanation: A flyover is curved; motion along a curved path is curvilinear motion.

4. **Correct Answer : A) 100 m**

Explanation: One round trip = 10 m + 10 m = 20 m

Total distance = $20 \times 5 = 100$m

5. **Correct Answer : Correct Answer : C) 0 m**

Explanation: Bird returns to same treetop position.

Displacement = Final position − Initial position = 0

6. **Correct Answer : B**

7. **Correct Answer : B) Less than or equal to**

Explanation:

Average velocity $= \frac{\text{Displacement}}{\text{Time}}$

Average speed $= \frac{\text{Distance}}{\text{Time}}$

Since Displacement ⩽ Distance, ratio ⩽ 1.

8. **Correct Answer : A) 20 m/s**

Explanation: $v = u + at$

$v = 0 + 4 \times 5 = 20$ m/s

9. **Correct Answer : A) Periodic**

Explanation: Motion repeats at equal intervals of time.

10. **Correct Answer : D) Oscillatory**

Explanation: Oscillatory motion repeats between two fixed positions.

11. **Correct Answer : C) Inner walls of train**

Explanation: With respect to train walls, position does not change ⇒ at rest.

12. **Correct Answer : D) Oscillatory motion**

Explanation: Pendulum moves to and fro about mean position.

13. **Correct Answer : B) 0**

Explanation: Round trip ⇒ final position = initial position.

Displacement = 0

14. **Correct Answer : A**

15. **Correct Answer : A) 3 m**

Explanation: Net displacement = 8 − 5 = 3 m (towards west)

16. **Correct Answer : Explanation:**

$$v = \frac{d}{t} = \frac{200}{2} = 100 \text{ m/s west}$$

17. Correct Answer : C) 50 kmph

Explanation: 120 minutes = 2 hours

$$\text{Average speed} = \frac{100}{2} = 50 \text{ km/h}$$

18. Correct Answer : B) 20 minutes

Explanation: $\text{Time} = \frac{\text{Distance}}{\text{Speed}} = \frac{20}{60}$ h

$$= \frac{1}{3} \text{ h} = 20 \text{ minutes}$$

19. Correct Answer : A) 100 m

Explanation: Displacement = Velocity × Time

$$s = 5 \times 20 = 100 \text{ m}$$

20. Correct Answer : D) 4 m/s

Explanation: $\Delta v = a \times t$

$$\Delta v = 2 \times 2 = 4 \text{ m/s}$$

***21.* Correct Answer : C) −3 m/s²**

Explanation: $a = \frac{v-u}{t}$

$$a = \frac{10 - 25}{5} = \frac{-15}{5} = -3 \text{ m/s}^2$$

Negative sign indicates retardation.

22. Correct Answer : A) Uniform velocity

Explanation: Straight-line displacement–time graph ⇒ constant slope.

$$\text{Velocity} = \frac{\text{Displacement}}{\text{Time}} = \text{constant}$$

ONE OR MORE THAN ONE OPTION CORRECT TYPE

23. A) Rotatory motion, B) Translatory motion

Explanation: A rolling ball rotates about its axis and also moves forward.
Rolling motion = Rotatory + Translatory motion.

24. Correct Answer : A) Spinning a top, B) Windmill, C) Fan, D) Wheels of a car

Explanation: Rotatory motion = motion about a fixed axis. All given objects rotate about an axis.

25. Correct Answer : A), B)

Explanation:
A) True — If displacement ≠ 0, distance must be covered.
B) True — In round trip, displacement = 0.
C) False — Distance ⩾ displacement (can be equal).
D) False — Distance can never be less than displacement.

26. Correct Answer : A), B)

Explanation:
A) True — Velocity has magnitude and direction (vector).
B) True —

$$\text{Velocity} = \frac{\text{Displacement}}{\text{Time}}$$

C) False — Displacement can be zero, so velocity can be zero.
D) False — Unit of displacement is metre (m).

27. Correct Answer : A), B), C), D)

Explanation:
Acceleration = Change in velocity ÷ Time

$$a = \frac{\Delta v}{t}$$

Velocity changes if magnitude changes or direction changes or both.
All given statements describe valid causes of acceleration.

ASSERTION AND REASON TYPE

28. Correct Answer : A) Both A and R are true and R is the correct explanation of A

Explanation: Distance is a scalar quantity and equals total path length, which cannot be negative; hence always positive.

29. Correct Answer : A) Both A and R are true and R is the correct explanation of A

Explanation: In circular motion (fan), direction continuously changes, so velocity (vector) changes even if speed is constant.

30. Correct Answer : A) Both A and R are true and R is the correct explanation of A

Explanation: Uniform speed means equal distances in equal time → distance ∝ time, so graph is a straight line.

31. Correct Answer : A) Both A and R are true and R is the correct explanation of A

Explanation: Displacement depends only on initial and final positions; if both are same, displacement = 0 even if distance > 0.

MATRIX MATCHING TYPE

32. a–R, b–S, c–Q, d–P

Explanation: Pendulum → Oscillatory motion (to and fro).
Car on straight road → Rectilinear motion.
Spinning top → Rotatory motion.
Bus on flyover → Curvilinear motion.

33. Correct Answer : a–Q, b–R, c–S, d–P

Explanation: Distance → Total path travelled.
Displacement → Shortest distance between two points.
Speed → Distance per unit time

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

Velocity → Has magnitude and direction.

34. Correct Answer : C) 25 m

Explanation: Distance $= 10 + 15 = 25$ m

35. Correct Answer : D) 5 m west

Explanation: Net displacement = 15 m west – 10 m east = 5 m west.

36. Correct Answer : B) 5 m/s

Explanation: Speed $= \frac{\text{Total distance}}{\text{Time}} = \frac{25}{5} =$ 5 m/s

37. Correct Answer : C) 324 km south

Explanation: Displacement equals distance in straight-line motion with direction south.

38. Correct Answer : A) 972

Explanation: 20 minutes = $\frac{1}{3}$hour

$$\text{Velocity} = \frac{324}{\frac{1}{3}} = 972 \text{ km/h}$$

39. Correct Answer : C) 60 km/h east

Explanation: Velocity $= \frac{90}{1.5} = 60$ km/h east

INTEGER TYPE

40. Correct Answer : 6 km/h

Explanation: Total distance = 12 + 18 = 30 km
Total time = 2 + 3 = 5 h

$$\text{Average speed} = \frac{30}{5} = 6 \text{ km/h}$$

41. Correct Answer : 4 m/s²

Explanation: $a = \frac{v-u}{t} = \frac{24-0}{6} = 4 \text{ m/s}^2$

42. Correct Answer : 5 m

Explanation: Net displacement = 8 − 3 = 5 m (east)
Magnitude = 5 m

PREVIOUS YEAR QUESTIONS

1. Correct Answer : Correct Option: C [Sunil has run a greater distance but his displacement is lesser than Rohit's]

Explanation: Distance: Sunil traveled to the end and more of the return leg, so his total path is longer.

Displacement: This is the straight-line distance from the origin. Since Sunil is further along his return path, he is physically closer to the origin than Rohit.

2. Correct Answer : C [14 m and 10 m]

Explanation: Distance $= 8$ m $+ 6$ m $=$ 14 m
Displacement $= \sqrt{8^2 + 6^2} = \sqrt{64 + 36} = \sqrt{100} = 10$ m

3. Correct Answer : B [An aero plane flying at a speed of 600 km/h]

Explanation: Uniform motion requires constant speed in a straight line. Options A, C, and D involve changing speeds.

4. Correct Answer : D [18 m/s]

Explanation: Time interval between images

$$= \frac{1}{12} \text{ s}$$

Total time

$$t = 4 \times \left(\frac{1}{12}\right) = \frac{1}{3} \text{ s}$$

Speed

$$= \frac{\text{Distance}}{\text{Time}} = \frac{6.0}{1/3} = 18 \text{ m/s}$$

5. Correct Answer : B [10–15 seconds]

Explanation: Speed $= \frac{20}{5} = 4$ m/s

6. Correct Answer : IF IN TEXTBOOK PRINT MISTAKE CHECK ONCE (6 AND 7 IS ONLY ONE MCQ SO TOTAL 8 MCQS IN PYQ NOT 9)

Correct Option: C [P-(ii), Q-(iii), R-(i)]

Explanation: P: Acceleration

$$a = \frac{\Delta v}{\Delta t}$$

At 20 s, $v = 40$. So,

$$a = \frac{40 - 0}{20} = 2 \text{ m/s}^2$$

Q: For the first 15 s, motion is uniform acceleration.

Distance = Area under $v - t$graph

$$= \frac{1}{2} \times 15 \times 30 = 225 \text{ m}$$

R: Distance for 30 s

$$= \text{Area of triangle (0–20 s)} + \text{Area of rectangle (20–30 s)}$$
$$= \left(\frac{1}{2} \times 20 \times 40\right) + (10 \times 40)$$
$$= 400 + 400 = 800 \text{ m}$$

7. Correct Answer : B [0.083 m/s]

Explanation: Average speed

$$= \frac{\text{Total Distance}}{\text{Total Time}}$$

Total time

$$4 \text{ minutes} = 240 \text{ seconds}$$

$$\text{Average speed} = \frac{20}{240} \approx 0.083 \text{ m/s}$$

8. Correct Answer : A [I and II only]

Explanation:

I. Cycle starts moving at 5 seconds. From

$$t = 0 \text{ to } t = 5 \text{ seconds}$$

the graph is a horizontal line at a distance of 2 m

This means the distance is not changing, so the cycle is stationary. At exactly

5 seconds

the line starts to slope upwards, indicating that the distance is beginning to change.

II: $\frac{9-3}{12-9} = 2 \text{ m/s}$

III: Average speed $= \frac{\text{Total Distance}}{\text{Total Time}} = \frac{9 \text{ m}}{12 \text{ s}} =$ 0.75 m/s

3. Heat

Solutions

WORKSHEET – 1

1. **Correct Answer : A) Hot, cold**

Explanation: Heat transfer always occurs from a body at higher temperature to a body at lower temperature until thermal equilibrium is reached (Based on Second Law of Thermodynamics).

2. **Correct Answer : C) It does not contain mercury**

Explanation: Mercury is toxic in nature; digital thermometers avoid mercury, making them safer for domestic use.

3. **Correct Answer : B) Scalar**

Explanation: Temperature has only magnitude and no direction ⇒ Scalar Quantity.

4. **Correct Answer : A) Laboratory thermometer**

Explanation: Laboratory thermometer is used to measure temperature of liquids and surroundings in labs and places.

5. **Correct Answer : C) Bulb**

Explanation: The bulb contains mercury which expands on heating; hence it must be in contact with the object for accurate measurement.

6. **Correct Answer : C) 212**

Explanation: Boiling point of water on Fahrenheit scale = **212°F** (Standard fixed point).

7. **Correct Answer : B) 180**

Explanation: No. of divisions = UFP – LFP = 212°F – 32°F = **180**

8. **Correct Answer : 12.5°C**

Explanation: Formula:$°C = \frac{5}{4}R$

$$°C = \frac{5}{4} \times 10 = 12.5°C$$

9. **Correct Answer : C) 312.4K**

Explanation: $K = \frac{5}{9}(F - 32) + 273$

$$K = \frac{5}{9}(103 - 32) + 273 = 39.4 + 273$$
$$= 312.4K$$

10. **Correct Answer : C) 131°F**

Explanation: $F = \frac{9}{5}C + 32$

$$F = \frac{9}{5} \times 55 + 32 = 99 + 32 = 131°F$$

WORKSHEET-2

1. **Correct Answer : B) Heat flows from spoon to water by conduction**

Explanation: Heat always flows from higher temperature to lower temperature through direct contact in solids ⇒ Conduction.

2. **Correct Answer : C) Heat transfer takes place due to movement of particles**

Explanation: In liquids, heat transfer occurs by convection involving bulk movement of fluid particles.

3. **Correct Answer : C) Feeling warm under sunlight**

Explanation: Heat from the sun reaches us through vacuum by radiation (no medium required).

4. **Correct Answer : C) Land heats faster and air above it rises**

Explanation: Land gets heated quickly; warm air rises creating low pressure, so cooler air from sea moves toward land (Sea breeze).

5. **Correct Answer : C) Is a good conductor and transfers heat quickly**

Explanation: Copper has high thermal conductivity, allowing rapid heat transfer by conduction.

6. **C) Black absorbs more thermal radiation**

Explanation: Black surfaces are good absorbers of radiant heat energy compared to white surfaces.

7. **Correct Answer : C) It requires movement of medium particles**

Explanation: Convection occurs due to actual movement of particles in fluids.

8. **Correct Answer : C) Air above sea becomes warmer and rises**

Explanation: At night, land cools faster than sea; warm air over sea rises and cooler air from land moves toward sea (Land breeze).

9. **Correct Answer : B) Prevent conduction and convection**

Explanation: Vacuum eliminates material medium, thus preventing conduction and convection heat losses.

10. **Correct Answer : C) Prevent heat transfer by radiation**

Explanation: Shiny inner walls reflect radiant heat back, reducing heat loss by radiation.

WORKSHEET-3

1. **Correct Answer :B) Mercury expands on heating**

Explanation: On heating, mercury expands causing the rise in level inside thermometer capillary.

2. **Correct Answer : B) Change in temperature**

Explanation: $Q = mc\Delta T \Rightarrow \Delta T = \frac{Q}{mc}$

For same material (same c) and same heat Q, heavier mass (m) ⇒ smaller temperature change.

3. **Correct Answer : C) Changing the state of ice into water**

Explanation: Heat supplied is used as latent heat of fusion to change state at constant temperature.

4. **Correct Answer : B) Evaporation**

Explanation: Evaporation is a slow surface phenomenon occurring at all temperatures below boiling point.

5. **Correct Answer : C) Sublimation**

Explanation: Direct change from solid to gas without passing through liquid state is sublimation.

6. **Correct Answer : B) Deposition**

Explanation: Direct conversion of gas into solid without becoming liquid is deposition.

7. **Correct Answer : C) Increase the distance between molecules**

Explanation: During change of state, latent heat increases intermolecular space without raising temperature.

8. **Correct Answer : C) Linear expansion**

Explanation: Increase in length of a solid due to heating is called linear expansion.

9. **Correct Answer : C) Transpiration**

Explanation: Loss of water vapour through stomata in plant leaves is transpiration.

10. **Correct Answer : B) Infiltration**

Explanation: Seepage of rainwater into ground forming groundwater is infiltration.

11. **Correct Answer : C) Infiltration**

Explanation: Downward movement of water through soil and rocks is infiltration.

12. **Correct Answer : C) Gravel**

Explanation: Gravel has larger pore spaces ⇒ higher infiltration rate.

COMPETITIVE CORNER

1. **Correct Answer :A) Kelvin**

Explanation: SI unit of temperature is Kelvin (K).

2. **Correct Answer : C) Temperature**

Explanation: Degree of hotness or coldness of a body is measured by temperature.

3. **Correct Answer : D) Both A and C**

Explanation: On heating, volume increases and density decreases due to expansion.

4. Correct Answer : Reaumur

5. Correct Answer :A) 273 K

Explanation: Melting point of ice = 0°C

$$K = °C + 273 = 273K$$

6. Correct Answer :D) −273°C

Explanation: Absolute zero = 0K

$$°C = K - 273 = -273°C$$

7. Correct Answer : A) 37.77°C

Explanation: $°C = \frac{5}{9}(F - 32)$

$$°C = \frac{5}{9}(100 - 32) = 37.77°C$$

8. correct answer is D)

9. **Correct Answer :**

Explanation: $T = \frac{(\theta - LFP)}{(UFP - LFP)} \times 100$

$$T = \frac{(65 - (-4))}{(92 - (-4))} \times 100 = \frac{69}{96} \times 100$$
$$= 71.88°C$$

Correct answer = 71.88°C (Option A)

10. **Correct Answer :D) 43.33°C**

Explanation:$°C = \frac{5}{9}(110 - 32) = \frac{5}{9} \times 78 = 43.33°C$

11. Correct Answer :C) Good conductors of heat

Explanation: Substances allowing easy heat flow are good conductors.

12. Correct Answer : C) Radiation

Explanation: Radiation is fastest as it does not require medium.

13. Correct Answer : B) Convection

Explanation: Boiling of entire liquid occurs due to convection currents.

14. Correct Answer : A) Thermal convection

Explanation: Heat transfer with actual movement of molecules occurs in convection.

15. Correct Answer : C

Explanation: $T = \frac{(\theta - LFP)}{(UFP - LFP)} \times 100$

$$T = \frac{(39-5)}{(95-5)} \times 100 = \frac{34}{90} \times 100 = 37.78°C$$

16. **Correct answer : = 37.78°C (Option C)**
16: A) 59.43°C

Explanation: $T = \frac{(60-(-3))}{(103-(-3))} \times 100 = \frac{63}{106} \times 100 = 59.43°C$

17. Correct Answer : B) 1.8°F

Explanation: $\Delta F = \frac{9}{5}\Delta C = \frac{9}{5} \times 1 = 1.8°F$

18. C) 50°C

Explanation: $T = \frac{(50-0)}{(100-0)} \times (105 - (-5)) + (-5) = \frac{50}{100} \times 110 - 5 = 50°C$

19. Correct Answer : B) 37°C

Explanation: Normal human body temperature = 37°C.

20. Correct Answer : C) Radiation

Explanation: Heat from heater reaches body without medium through radiation.

21. Correct Answer : A) Sea breeze

Explanation: Air moving from sea towards land is called sea breeze.

22. Correct Answer : B) Air

Explanation: Air is a poor conductor ⇒ minimum heat transfer by conduction.

23. Correct Answer : A) Melting point

Explanation: Fixed temperature at which solid changes to liquid is melting point.

24. Correct Answer : B) Heat is absorbed

Explanation: During melting, latent heat of fusion is absorbed.

25. Correct Answer : B) Boiling point

Explanation: Fixed temperature at which liquid changes into gas is boiling point.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

26. Correct Answer : A) 98.4°F, D) 36.9°C

Explanation: Normal human body temperature ≈ 37°C = 98.6°F (acceptable clinical range includes 36.9°C and 98.4°F).

27. Correct Answer : A) SI unit of temperature is Kelvin, B) Temperature is a scalar quantity, C) SI unit of heat is joule (J), D) 1 Cal = 4.2J

Explanation: SI unit of temperature = Kelvin; Heat is energy measured in Joule;

$$1 \text{ Cal} = 4.2 \text{ J}$$

28. Correct Answer : B) Medium is required for transfer of heat by conduction, C) Radiation is possible in vacuum, D) Medium is required for the transfer of heat by convection

Explanation: Conduction and convection require medium; radiation does not require medium and can occur in vacuum.

29. Correct Answer : B) A medium is necessary for convection, C) Convection involves movement of particles

Explanation: Convection occurs in fluids with actual movement of particles and requires a material medium.

30. Correct Answer : A) Melting of ice, B) Boiling of water, C) Sublimation of a solid

Explanation: All involve change of state due to latent heat; linear expansion is dimensional change only.

ASSERTION AND REASON TYPE

31. Correct Answer : A) Both A and R are true and R is the correct explanation of A

Explanation: Laboratory thermometer lacks kink, hence mercury falls back immediately and cannot hold human body temperature reading.

32. Correct Answer : A) Both A and R are true and R is the correct explanation of A

Explanation: Land heats faster than sea → air above land rises → cooler air from sea moves to land (Sea breeze).

33. Correct Answer : A) Both A and R are true and R is the correct explanation of A

Explanation: During melting, latent heat of fusion is used to change state at constant temperature.

34. Correct Answer : A) Both A and R are true and R is the correct explanation of A

Explanation: Black surfaces are good absorbers of thermal radiation, increasing heating efficiency.

MATRIX MATCHING TYPE

35. Correct Answer : a–Q, b–P, c–S, d–R

Explanation:
Celsius → LFP = 0°C
Fahrenheit → UFP = 212°F
Kelvin → LFP = 273K
Reaumur → UFP = 80°R

36. Correct Answer : a–Q, b–P, c–S, d–R

Explanation: Solid→Liquid = Melting
Liquid→Gas = Evaporation
Solid→Gas = Sublimation
Gas→Liquid = Condensation

COMPREHENSION TYPE

37. Correct Answer : D) 100°C

Explanation: Boiling point of water at standard atmospheric pressure = 100°C.

38. Correct Answer : A) 32°F

Explanation: $F = \frac{9}{5}C + 32$

$$F = \frac{9}{5} \times 0 + 32 = 32°F$$

39. **Correct Answer : B) 100°C**

Explanation: $C = \frac{5}{9}(F - 32)$

$$C = \frac{5}{9}(212 - 32) = 100°C$$

40. **Correct Answer :A) Polishing the surface**
Explanation: Polished surfaces are poor emitters and absorbers of radiation, reducing heat loss.

41. Correct Answer : D) Conduction and convection
Explanation: Vacuum eliminates medium, preventing conduction and convection losses.

42. Correct Answer : C) Thermos flask
Explanation: Thermos minimizes heat loss by conduction, convection (vacuum) and radiation (shiny walls).

INTEGER TYPE

43. Correct Answer : 20°C

Explanation: $C = \frac{5}{9}(68 - 32) = \frac{5}{9} \times 36 = 20°C$

44. Correct Answer : 16°R

Explanation: $R = \frac{4}{5}C = \frac{4}{5} \times 20 = 16°R$

45. Correct Answer : 81°F

Explanation: $\Delta F = \frac{9}{5}\Delta C = \frac{9}{5} \times 45 = 81°F$

PREVIOUS YEAR QUESTIONS

1. Correct Answer : C) Rod Y being a good conductor of heat, conducts away the heat given to the paper

Explanation: A good conductor transfers heat away from the heated region quickly by conduction, preventing the paper from reaching its ignition temperature.

2. Correct Answer : D) II, III and IV only

Explanation:
I – False (Thermometer should be held by stem, not bulb)
II – True (Jerks bring mercury below 35°C before use)
III – True (Black absorbs more heat in winter)
IV – True (Jute is poor conductor, reduces heat gain)

3. Correct Answer : D) Greater sensitivity

Explanation: Sensitivity ∝ expansion per °C; liquid with higher coefficient of expansion gives larger rise for same temperature change.

4. Correct Answer : A) I only

Explanation: During melting at 0°C, latent heat is absorbed to break intermolecular bonds without change in temperature.

5. Correct Answer : B) (i) and (ii) only

Explanation: Heat from flame is conducted to water; water absorbs heat and its boiling point (100°C) is lower than ignition temperature of paper, preventing it from burning.

6. Correct Answer : C) 1.0°C lower than the true temperature

Explanation:
Thermometer shows:

Melting ice → reads −1°C instead of 0°C
Steam → reads 99°C instead of 100°C

So, it always shows 1°C less than the actual temperature.

Therefore, measured temperature is 1°C lower than the true temperature.

7. Correct Answer : B) (i) and (iv) only

Explanation: On heating, both steel bars expand causing the gap between them to decrease; metals are good conductors of heat.
(ii) is false because matter expands on heating.
(iii) is false because both bars expand in length.

8. Correct Answer : (ii), (iii) and (iv)

Explanation: Convection in liquids occurs when there is temperature difference causing movement of particles.
(ii) Electric heater heats water from inside → convection currents form
(iii) Ice cools surrounding water → denser water sinks and warmer water rises
(iv) Candle heats water from below → convection currents form

9. Correct Answer : C

Explanation: On heating from below, water near the flame becomes hot and rises; cooler water from top sinks forming a clockwise convection current

4. Light – Shadows and Reflection

Solutions

WORKSHEET – 1

1. **Correct Answer: C) Bounces back into the same**

   medium after striking a surface
   Reflection of light occurs when light strikes a surface and returns back into the same medium. This usually happens with mirrors and shiny surfaces.

2. **Correct Answer: B) Convergent beam**

   When light rays coming from different directions meet at a single point, they form a convergent beam. This is commonly seen when a concave mirror focuses light.

3. **Correct Answer: B) A beam**

   A beam is a collection of many light rays travelling together. For example, sunlight entering through a window forms a beam.

4. **Correct Answer: A) A bulb**

   An extended source of light has a noticeable size and emits light from many points. A bulb is an example because it spreads light from its entire surface.

5. **Correct Answer: C) Wall**

   Diffused reflection occurs when light falls on a rough surface like a wall. The reflected rays scatter in different directions.

6. **Correct Answer: C) Glancing angle of incidence**

   The angle between the incident ray and the mirror surface is called the glancing angle of incidence. It is complementary to the angle of incidence.

7. **Correct Answer: C) Because light cannot pass**

   through opaque objects
   A shadow is formed when an opaque object blocks light. Since light cannot pass through the object, a dark region appears behind it.

8. **Correct Answer: C) Angle of incidence**

   The angle of incidence is the angle between the incident ray and the normal at the point of incidence. It is an important part of the laws of reflection.

9. **Correct Answer: B) Controlling the puppets and the light**

   In shadow puppetry, movement of puppets between the light source and screen creates moving shadows. Proper control of light and puppets makes characters appear to move.

10. **Correct Answer: B) It decreases**

    When light enters water from vacuum, its speed decreases because water is optically denser than vacuum. Light travels fastest in vacuum.

WORKSHEET-2

1. **Correct Answer: C) Image**

   When reflected rays actually meet (real image) or appear to meet (virtual image), the meeting point is called an image.
   It is the position where the object seems to be formed after reflection.

2. **Correct Answer: C) A real image is formed when**

   rays actually meet, whereas a virtual image is formed when rays appear to meet.
   In a real image, light rays physically converge at a point and can be obtained on a screen.
   In a virtual image, rays only appear to meet and cannot be captured on a screen.

3. **Correct Answer: C) Lateral inversion**

In a plane mirror, the left side of an object appears as the right side and vice versa. This sideways reversal is called lateral inversion.

4. **Correct Answer: B) 3**

When two plane mirrors are placed at 90°, the number of images formed is given by: Number of images = 360°/θ − 1 = 360/90 − 1 = 4 − 1 = 3.

5. **Correct Answer: C) 360 / θ − 1**

The number of images formed by two mirrors inclined at an angle θ is: Number of images = 360/θ − 1 (when 360/θ is an integer).

6. **Correct Answer: D) In dressing tables**

A periscope is used in submarines, trenches, and tanks to see objects not directly visible. Dressing tables use plane mirrors, not periscopes.

7. **Correct Answer: C) Light rays actually meet at a point**

In a pinhole camera, light rays pass through a small hole and actually converge on the screen.
Since rays really meet, the image formed is real.

8. **Correct Answer: B) 60°**

In a kaleidoscope, three plane mirrors are arranged at 60° to each other.
This arrangement produces multiple symmetrical patterns.

9. **Correct Answer: B) Multiple reflection of light**

A kaleidoscope works on the principle of multiple reflection of light.
Light reflects repeatedly between mirrors, forming beautiful symmetrical patterns.

10. **Correct Answer: B) It is inverted**

An image formed by a plane mirror is erect (upright), not inverted.
It is virtual, same in size as the object, and shows lateral inversion.

WORKSHEET-3

1. **Correct Answer: C) Principal axis**

The principal axis is the imaginary straight line joining the pole (P) and the centre of curvature (C). It is the main reference line for studying image formation in spherical mirrors.

2. **Correct Answer: B) Behind the mirror**

In a convex mirror, the reflecting surface bulges outward. Hence, its centre of curvature lies behind the mirror in the virtual region.

3. **Correct Answer: B) At F**

When light rays come from infinity, they are parallel to the principal axis. After reflection from a convex mirror, they appear to diverge from the focus (F), so the image forms at F behind the mirror.

4. **Correct Answer: B) Virtual and erect**

A convex mirror always forms a virtual image behind the mirror. The image is erect and diminished regardless of the object position.

5. **Correct Answer: B) Beyond C**

For a concave mirror, when the object is placed between F and C, the image forms beyond C. The image is real, inverted, and magnified.

6. **Correct Answer: B) Real, inverted and same size**

If the object is placed at the centre of curvature (C), the image is formed at C itself. It is real, inverted, and of the same size as the object.

7. **Correct Answer: C) Gives wider field of view**

A convex mirror diverges light rays, allowing a larger area to be seen. This wider field of view makes it suitable for rear-view mirrors.

8. **Correct Answer: C) Virtual and erect**

A convex mirror always produces a virtual, erect, and diminished image for any object position.

9. Correct Answer: D) Between F and P

When the object is placed between the focus (F) and pole (P) of a concave mirror, the image formed is virtual, erect, and enlarged behind the mirror.

10. Correct Answer: C) Retrace its path after reflection

A ray passing through the centre of curvature strikes the mirror normally (perpendicular). Hence, it reflects back along the same path.

SINGLE OPTION CORRECT TYPE

1. Correct Answer : is A) 3×10^8 m/s

The speed of light in vacuum is 3×10^8 metres per second. It is the fastest speed known in nature

2. Correct Answer : is B) Transparent medium

A transparent medium allows most of the light to pass through it. Objects can be seen clearly through it.

3. Correct Answer : is C) Point of incidence

It is the exact point where the incident ray strikes the reflecting surface. Reflection begins from this point.

4. Correct Answer : is B) Same plane

The incident ray, reflected ray, and normal always lie in the same plane. This is the first law of reflection.

5. Correct Answer : is B) Reflected ray

The ray that bounces back into the same medium after striking a surface is called the reflected ray.

6. Correct Answer : is A) Parallel reflected rays

(Regular reflection)
In regular reflection, parallel incident rays remain parallel after reflection. This happens on smooth surfaces.

7. Correct Answer : is B) Reflection of light

Reflection of light is the bouncing back of light into the same medium after striking a surface.

8. Correct Answer : is D) Glancing angle of reflection

It is the angle between the reflected ray and the mirror surface. It is complementary to the angle of reflection.

9. Correct Answer : is A) 45°

According to the law of reflection, angle of incidence equals angle of reflection. Hence, 45°.

10. Correct Answer : is A) 10°

Glancing angle = 90° – angle of incidence. So, 90° – 80° = 10°.

11. Correct Answer : B) Convex

A convex mirror provides a wide field of view and forms erect, diminished images. This makes it suitable for rear-view mirrors in vehicles.

12. Correct Answer : A) Pole

The pole is the midpoint of the reflecting surface of a spherical mirror. It lies on the principal axis.

13. Correct Answer : A) 32 cm

For spherical mirrors, $R = 2f$.
So, $R = 2 \times 16 = 32$cm.

14. Correct Answer : B) Convex

In a convex mirror, the reflecting surface bulges outward. The outer curved surface reflects light.

15. Correct Answer : B) Real

A concave mirror converges parallel rays to a point in front of the mirror. Therefore, its principal focus is real.

16. Correct Answer : A) Extremely diminished

When the object is at infinity, parallel rays appear to meet at the focus behind the convex mirror. The image formed is highly diminished.

17. Correct Answer : B) Virtual, magnified

Since the object is placed within the focal length (5 cm < 15 cm), the image formed is virtual, erect, and magnified.

18. Correct Answer : A) Polished surface

A smooth polished surface produces regular reflection and forms clear images. Rough surfaces scatter light and cannot act as mirrors.

19. Correct Answer: C) 50°

By the law of reflection, the angle of reflection is always equal to the angle of incidence.
Since the angle of incidence is 50°, the angle of reflection is also 50°.

20. Correct Answer: B) 60°

Angle of incidence is measured from the normal, not from the mirror surface.
If the ray makes 30° with the surface, then with the normal it makes $90° - 30° = 60°$.

21. Correct Answer: A) 90°

Glancing angle is the angle between the incident ray and the mirror surface.
Glancing angle = $90° - i$. If $i = 0°$, glancing angle = 90°.

22. Correct Answer: A) Concave mirror

A concave mirror can reflect light from a point source placed at its focus into a parallel beam.
This property is used in torches and searchlights.

23. Correct Answer: A) Under no circumstances

A convex mirror always forms a virtual, erect, and diminished image.
It never produces an inverted image for any object position.

24. Radius of curvature (Correct Answer: C) 40 cm

PF represents the focal length (f) = 20 cm.
For a spherical mirror, $R = 2f$.
So, $R = 2 \times 20 = 40$cm.

25. X denotes (Correct Answer: A) Centre of curvature

The ray shown passes through point X and reflects back along the same path.
A ray passing through the centre of curvature retraces its path after reflection.
Hence, X is the centre of curvature.

26. For a convex mirror (Correct Answer: B) Appears to pass through the focus

In a convex mirror, a ray directed towards the focus (behind the mirror) reflects parallel to the principal axis.
Since the focus is virtual, the ray only appears to pass through it.

27. Correct Answer: A) 0°

When a ray is incident along the normal, the angle of incidence is 0°.
By the law of reflection, angle of reflection = angle of incidence.
Hence, the angle of reflection is also 0°.

28. Correct Answer: C) 1 : 1

According to the law of reflection, angle of incidence is always equal to angle of reflection.
Therefore, their ratio is 1 : 1.

29. Correct Answer: C) A convex mirror

A convex mirror always forms a virtual, erect, and diminished image for a real object.
Plane mirrors form virtual but same-sized images, while concave mirrors may form magnified images.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

30. Correct Answer: A and B

Explanation:
According to the laws of reflection:

1. Angle of incidence = Angle of reflection.
2. Incident ray, reflected ray and normal lie in the same plane.
   Hence A and B are correct.

31. Correct Answer: B and C

Explanation:
Angle of reflection = Angle of incidence = 70° → B is correct.

Glancing angle = 90° − 70° = 20° → C is correct.

32. Correct Answer: A, B and D

Explanation:
Regular reflection occurs on smooth surfaces like plane mirrors, highly polished metals and still water.
Road is rough, so it gives irregular reflection.

33. Correct Answer: A, B, C and D

Explanation:
For every angle of incidence, angle of reflection is equal to it.
So at 0°, 40°, 35°, and 15° — they are equal.

34. Correct Answer: A and B

Explanation:
Plane mirror and convex mirror always form virtual and erect images for all object positions.
Concave mirror does not.

ASSERTION AND REASON TYPE

35. Correct Answer: A

Both A and R are true and R is the correct explanation of A.
When an object is placed beyond C in a concave mirror, the image forms between F and C. The image is real and inverted, which is a characteristic of image formation in this position and explains the assertion.

36. Correct Answer: B

Both A and R are true but R is not the correct explanation of A.
Jewellery shops use plane mirrors to make the shop look bigger because mirrors create an illusion of extended space. Multiple reflections may occur, but that is not the main reason for the enlarged appearance.

37. Correct Answer: A

Both A and R are true and R is the correct explanation of A.
Regular reflection occurs on smooth surfaces like a plane mirror. A smooth surface reflects parallel rays in one direction, producing clear and distinct images.

38. Correct Answer: A

Both A and R are true and R is the correct explanation of A.
In a plane mirror, the image distance is equal to the object distance. Therefore, if you stand 2 m in front of the mirror, your image forms 2 m behind it.

MATRIX MATCHING TYPE

39. Correct Answer: a–S, b–P, c–R, d–Q

40. Correct Answer: a–Q, b–S, c–P, d–T

COMPREHENSION TYPE

41. Correct Answer: The angle of reflection is

$$\angle r = 80^\circ$$

Correct Answer: C) 80°

Explanation: Angle of reflection equals angle of incidence. Since angle of incidence is 80°, reflection is also 80°.

42. Correct Answer: The angle of incidence is

$$\angle i = 90^\circ - 10^\circ = 80^\circ$$

Correct Answer: A) 80°

Explanation: The given 10° is with the mirror surface. Angle of incidence is measured from the normal, so subtract from 90°.

43. Correct Answer: Angle of incidence + angle of reflection

$$80^\circ + 80^\circ = 160^\circ$$

Correct Answer: B) 160°

Explanation: Since both angles are equal (80° each), their sum is 160°.

44. Correct Answer: D) All of these

Letters U, V, and W are vertically symmetrical, so their mirror image looks the same.
Hence, they do not show lateral inversion in a plane mirror.

45. Correct Answer: B) Z

Letter Z is not vertically symmetrical, so its mirror image appears reversed.
Therefore, it shows lateral inversion in a plane mirror.

46. Correct Answer: D) All of these

A) In a mirror, left appears as right due to lateral inversion.
B) 'AMBULANCE' is written in reverse so drivers can read it correctly in mirrors.
C) The word 'Red' also appears laterally inverted in a plane mirror.

INTEGER TYPE

47. Correct Answer: 30 cm

For a spherical mirror, Radius of curvature (R) = 2 × Focal length (f).
So, R = 2 × 15 = 30 cm.

48. Correct Answer: 1

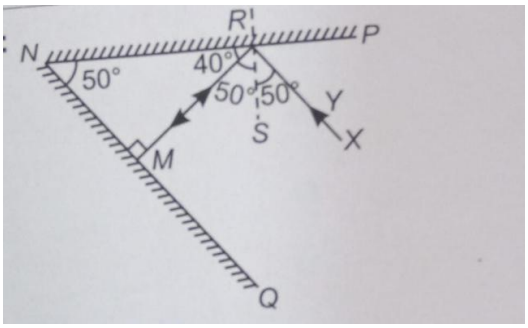
When the object is at the centre of curvature of a concave mirror, the image formed is of the same size.
Magnification (m) = 1.

49. Correct Answer: 3

Number of images formed by two plane mirrors inclined at 90° is given by
$n = \frac{360}{\theta} - 1 = \frac{360}{90} - 1 = 3.$

PREVIOUS YEAR QUESTIONS

1. Correct Answer: is B)



By law of reflection,
∠XRS = ∠MRS = 50°

(∵ ∠i = ∠r)

In ΔRNM,
∠MRN = 40°; ∠RNM = 50° (given)
∴ ∠RMN = 90°

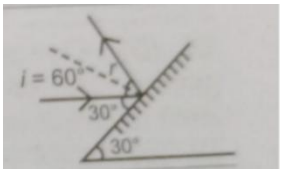
As the light ray is incident normally on the mirror Q, it will get retrace the same path. Hence, angle of reflection is 0°.

2. Correct Answer is B): The rays of light coming

from the object P only can reach the eye after reflected back by the mirror, so only object P is seen in the mirror.

3. Correct Answer: (B)

Angle of incidence,
$\angle i = 90° - 30°$
$= 60°$



By law of reflection,
$\angle r = \angle i = 60°$
Angle of incidence = 60°
Angle of reflection = 60°

4. Correct Answer: is (B) : The object at larger

distance from the mirror that is closer to the obstacle will disappear first. Thus R will disappear first.

5. Correct Answer: is (A) : In case of spherical

mirrors (concave or convex) the incident parallel rays after reflection may converge or diverge. It is only possible in case of plane mirror that the parallel incident rays reflected as a parallel beam.

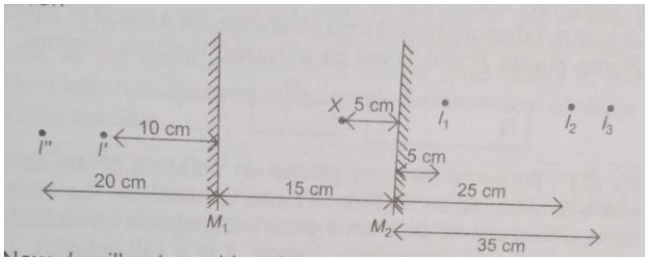
6. Correct Answer: is (B): Image (I_1) formed by M_2

will be at 5 cm behind the mirror.

Distance between mirror M_1 and object X is 10 cm.

So, image (I') formed by M_1 will be 10 cm behind M_1.

This image will act as object for M_2.
Now, distance between this object and M_2 = 10 + 15 = 25 cm.
Second image (I_2) formed by M_2 will be 25 cm behind the mirror.



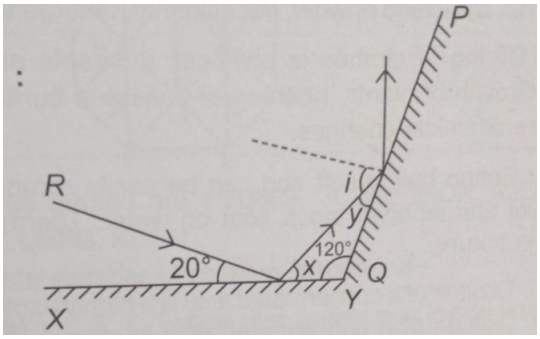
Now, I_1 will act as object for M_1 and the distance between I_1 and M_1 is 20 cm. So the image (I″) formed by M_1 will be 20 cm behind the mirror.

Image I″ will act as object for M_2 and distance between this object (I″) and M_2 = 20 + 15 = 35 cm.

Third image (I_3) formed by M_2 will be 35 cm behind M_2.

7. Correct Answer: is B)

Angle $x = 20°$



$$y = 180° - 120° - 20°$$

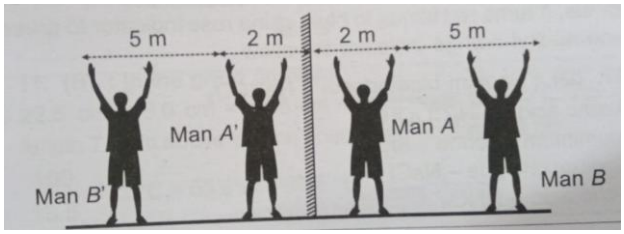
$$y = 40°$$

$$i = 90° - y$$

$$i = 90° - 40° = 50°$$

8. Correct Answer is D) : The image is at a same

distance as that of object infront of plane mirror,



So distance = 5 + 2 + 2 = 9 m

9. Correct Answer : is (D) : If the mirror is rotated

by an angle θ, the new reflected ray is rotated by an angle 2θ in same direction. So, it rotates 20° anticlockwise.

5. Electric Current and Its Effects

Solutions

WORKSHEET – 1

1. Correct Answer : A) Battery

Explanation: A battery is a combination of two or more cells connected in series or parallel to provide higher voltage/current.

2. Correct Answer : A) Chemical energy into Electrical energy

Explanation: A cell works on electrochemical reactions, converting chemical energy into electrical energy.

3. Correct Answer : A) Negative

Explanation: When ebonite is rubbed with cat's skin, it gains electrons and becomes negatively charged (electron transfer principle).

4. Correct Answer : C) Coulomb

Explanation: SI unit of electric charge is coulomb (C). $Q = It$.

5. Correct Answer : D) Ampere

Explanation: SI unit of electric current is ampere (A). $I = \frac{Q}{t}$.

6. Correct Answer :

Explanation: $I = \frac{Q}{t}$. One ampere is the current when 1 coulomb flows in 1 second.

7. Correct Answer: $V = \frac{W}{Q}$

Explanation: Potential difference is work done per unit charge. $V = \frac{W}{Q}$.

8. Correct Answer : B) Dilute sulphuric acid

Explanation: In a simple voltaic cell, electrolyte used is dilute H_2SO_4.

9. Correct Answer : A) Leclanché cell

Explanation: Dry cell is an improved version of Leclanché cell with paste electrolyte instead of liquid.

10. Correct Answer : C) Series

Explanation: When cells are connected end to end (positive to negative), it is series connection.

11. Correct Answer : C) Carbon plates

Explanation: In a bichromate cell, carbon acts as positive electrode and zinc as negative electrode.

12. Correct Answer : C) 2 V

Explanation: The emf of a bichromate cell is approximately 2 volts.

13. Correct Answer : C) Does not change its direction with time

Explanation: Direct current (DC) is defined as the flow of electric charge in a single, fixed direction. Unlike alternating current (AC), where the direction reverses periodically, DC maintains a constant polarity (positive to negative). The magnitude of DC may be constant or slightly varying, but the **direction remains unchanged**. Examples include batteries and cells. Hence, DC is characterized by unidirectional flow of charge.

14. Correct Answer : C) Frequency

Explanation: Frequency refers to the number of complete cycles (one full positive and negative variation) of alternating current (AC) completed in one second. It is denoted by **f** and measured in **Hertz (Hz)**.
Formula: $f = \frac{1}{T}$, where **T** is the time period of one cycle.
For example, in India, AC supply has a frequency of **50 Hz**, meaning 50 cycles occur every second.

WORKSHEET-2

1. **Correct Answer : B) Heating effect of current**

   Explanation: When current flows through the thin filament, it gets very hot and starts glowing. This happens due to the heating effect of current.
   Formula: $H = I^2Rt$

2. **Correct Answer : D) 3,000,000 Ω**

   Explanation: 1 mega ohm (MΩ) = 10,00,000 Ω.
   So, 3 MΩ = 3 × 10,00,000 = 30,00,000 Ω = 3,000,000 Ω.

3. **Correct Answer : B) Show electronic readings**

   Explanation: Digital meters show the reading directly in numbers, so it is easy to read and more accurate.

4. **Correct Answer : C) ISI mark**

   Explanation: The ISI mark tells us that the product is tested and safe to use.

5. **Correct Answer : B) Unequal**

   Explanation: In a parallel circuit, current divides into different paths. If resistances are different, current will also be different.
   Formula: $I = \frac{V}{R}$

6. **Correct Answer : B) Decreases**

   Explanation: When bulbs are connected in series, total resistance increases. So less current flows, and each bulb glows dimmer.
   Formula: $I = \frac{V}{R}$

7. **Correct Answer : C) Short circuit**
   Explanation: If too much current flows, wires become very hot and may cause a short circuit.

8. **Correct Answer : B) Parallel**
   Explanation: In houses, appliances are connected in parallel so each gets full voltage and works independently.

9. **Correct Answer : C) Melts the wire in the fuse**

   Explanation: If excess current flows, the fuse wire heats up and melts, breaking the circuit and protecting appliances.
   Formula: $H = I^2Rt$

10. **Correct Answer : C) Low resistance**

    Explanation: Copper allows current to pass easily because it has low resistance.

11. **Correct Answer : B) Fuse**

    Explanation: A fuse is a thin wire that melts when too much current flows.

12. **Correct Answer : B) Series**

    Explanation: Decorative bulbs are usually connected in series so that the voltage is shared among them.

13. **Correct Answer : C) Same**

    Explanation: In a series circuit, the same current flows through all bulbs, so identical bulbs glow with the same brightness.

WORKSHEET-3

1. **Correct Answer : D) Electric bell**

   Explanation: An electric bell uses an electromagnet to pull the hammer and make a ringing sound.

2. **Correct Answer : B) As long as current flows through it**

   Explanation: The magnetic field is present only when electricity is flowing. If current stops, magnetism also stops.

3. **Correct Answer : A) Increase**

   Explanation: When more current flows, the magnetic effect becomes stronger. (Simple idea: More current → Stronger magnet)

4. **Correct Answer : A) Electromagnet**

   Explanation: A magnet made by using electric current is called an electromagnet.

5. **Correct Answer : C) 1 watt**

   Explanation: Power means how fast energy is used.
   If 1 joule is used in 1 second:

$$P = \frac{E}{t} = \frac{1}{1} = 1\ W$$

6. **Correct Answer : C) Collision of electrons with atoms**

Explanation: Moving electrons hit atoms inside the wire. These collisions produce heat.

7. **Correct Answer : A) 0.3 kWh**

Explanation:
Energy = Power × Time
60 W = 0.06 kW
0.06 × 5 = 0.3 kWh

8. **Correct Answer : B) Rs. 20**

Explanation:
Total cost = Units × Cost per unit
4 × 5 = Rs. 20

9. **Correct Answer : B) Is a good insulator and good conductor of heat**

Explanation: Magnesium oxide does not allow electricity to pass but allows heat to pass safely.

10. **Correct Answer : C) Behaves like a magnet Explanation:**

Hans Christian Oersted discovered that a compass needle deflects near a current-carrying wire, proving that the wire generates a magnetic field.

11. **Correct Answer : B) 2 kWh**

Explanation:
1000 W = 1 kW
Energy = 1 × 2 = 2 kWh

12. **Correct Answer : B) 4 A**

Explanation: $P = VI$

$$I = \frac{880}{220} = 4\ A$$

13. **Correct Answer : B) Increase in current**

Explanation: When power increases at same voltage, current increases. More current produces more heat, so fuse melts.

14. **Correct Answer : B) Screw Explanation: The contact**

screw acts as the interrupter. When the electromagnet pulls the armature, contact with the screw acts to break the circuit, stopping the current and resetting the cycle (Make-and-Break mechanism).

15. **Correct Answer : B) Solder**

Explanation: Solder acts as the filler metal (alloy) melted to join wires or components in soldering.

COMPETITIVE CORNER

1. **Correct Answer : B) Electric cell**

Explanation: An electric cell is a small, portable source of electric current. Powerhouses and generators are large-scale sources.

2. **Correct Answer : A) Positive**

Explanation: When glass is rubbed with silk, electrons transfer from glass to silk. Loss of electrons makes the glass positively charged.

3. **Correct Answer : B) Electron**

Explanation: Electron carries negative charge (−e), proton is positive (+e), neutron is neutral.

4. **Correct Answer : D(Correct Answer: 1.6×10^{-19} C)**

Explanation: Magnitude of charge on electron is $e = 1.6 \times 10^{-19}$C.

5. **Correct Answer : D) The number of negative charges is**

equal to positive charges.
Explanation: A body is neutral when net charge is zero, i.e., Total positive charge = Total negative charge.

6. **Correct Answer : C) Electric current**

Explanation: Electric current is the rate of flow of charge.
Formula: $I = \frac{Q}{t}$

7. **Correct Answer : B (Correct Answer: $I = \frac{Q}{t}$)**

Explanation: Current equals charge divided by time.

8. **Correct Answer : B) Galvanometer**

Explanation: Galvanometer measures very small (sensitive) currents.

9. Correct Answer : B) Electric potential

Explanation: Electric potential is work done per unit charge.
Formula: $V = \frac{W}{Q}$

10. Correct Answer : D) Voltmeter

Explanation: Voltmeter measures potential difference across two points.

11. Correct Answer : C) Energy

Explanation: Chemical reactions inside a cell provide electrical energy.

12. Correct Answer : C

Explanation: Resistance is represented symbolically by R and in circuits by a zig-zag line.

13. Correct Answer : B) Ohm

Explanation: SI unit of resistance is Ohm (Ω).
Formula: $R = \frac{V}{I}$

14. C) 1.0 V (Approximate)

Explanation: A simple voltaic cell has emf ≈ 1.08 V, so 1.0 V is the closest correct option.

15. Correct Answer : B) Two
Explanation: An electric cell has two terminals: positive and negative.

16. Correct Answer : D) Chemical
Explanation: An electric cell stores chemical energy.

17. Correct Answer : D) Electrical
Explanation: A cell converts chemical energy into electrical energy.

18. Correct Answer : A: The standard symbol of a cell
consists of one long line (+) and one short line (–).

19. Correct Answer : B) Positive terminal
Explanation: In the symbol of a cell, the longer line represents the positive terminal.

20. Correct Answer : B) Parallel
Explanation: Connecting all positive terminals together and all negative terminals together forms a parallel connection.

21. Correct Answer : B

When more cells are connected in series in an electric circuit, the total potential difference (voltage) increases. According to Ohm's Law,

22. Correct Answer : C) Does not glow

Explanation: In series circuit, if one bulb fuses, the circuit breaks and all bulbs go off.

23. Correct Answer : A) Continue to glow

Explanation: In parallel circuit, each bulb has independent path, so others continue glowing.

24. Correct Answer : D) Miniature circuit breaker

Explanation: MCB stands for Miniature Circuit Breaker.

25. Correct Answer : 25: B) Filament

Explanation: The tungsten wire inside bulb that glows is called filament.

26. Correct Answer : B) Heated up

Explanation: Filament glows due to heating effect of current (incandescence).

27. Correct Answer : B) Alternating current (AC)

Explanation: Household electricity supply is AC (alternating direction periodically).

28. Correct Answer : B) Parallel

Explanation: Voltmeter has high resistance and is connected in parallel to measure potential difference.

29. Correct Answer : C) kWh meter

Explanation: Electrical energy consumption is measured in kilowatt-hour (kWh).
1 unit = 1 kWh

30. Correct Answer : B) High resistance

Explanation: Nichrome has high resistivity and high melting point, so it produces heat efficiently.
Formula: $H = I^2Rt$

31. Correct Answer : B) Current increases

Explanation: Strength of electromagnet is directly proportional to current.
Relation: $B \propto I$

ONE OR MORE THAN ONE CORRECT OPTION TYPE

32. Correct Answer : A) Ammeter, C) Galvanometer

Explanation: Ammeter measures electric current in circuits. Galvanometer measures very small currents. Voltmeter measures voltage, and odometer measures distance.

33. Correct Answer : A) w = vq, D) v = w/q

Explanation: Electric potential is defined as $V = \frac{W}{Q}$.
Rearranging gives $W = VQ$.
Option B (q = wv) and C (w = q/v) are incorrect.

34. Correct Answer : A) Fuse, D) MCB

Explanation: Fuse and MCB protect household circuits from overcurrent and short circuit. Resistor and ammeter are not protection devices.

35. Correct Answer : C) Ammonium chloride

Explanation: In a Leclanché cell, the electrolyte used is ammonium chloride (NH_4Cl) paste.

36. Correct Answer : C) Carbon

Explanation: In a simple Leclanché cell, the positive terminal (cathode) is the carbon rod; zinc acts as the negative terminal (anode).

ASSERTION AND REASON TYPE

37. Correct Answer : D) A is false but R is true

Explanation: Heat is produced due to the heating effect of electric current. It is clearly stated that the more the electrical resistance of a conductor, the more is the heat produced. Therefore, more heat is not produced in low resistance.
Heat is produced because free electrons collide with atoms and their kinetic energy appears in the form of heat. Hence, Assertion is false but Reason is true.

38. Correct Answer : C) A is true but R is false

Explanation: In series combination, the equivalent resistance is equal to the sum of individual resistances. Therefore, total resistance is greater than each individual resistance. However, resistances are added in **series**, not in parallel. Hence, Assertion is true but Reason is false.

39. Correct Answer : A) Both A and R are true and R is the correct explanation of A

Explanation: In parallel connection, all resistors are connected between the same two points. Therefore, the voltage across each resistor is the same. The Reason correctly explains the Assertion.

40. Correct Answer : A) Both A and R are true and R is the correct explanation of A

Explanation: An electric bell works due to repeated making and breaking of the circuit. If the circuit remains permanently closed, this making-and-breaking process stops, and the bell does not ring continuously. Hence, both Assertion and Reason are true, and Reason correctly explains Assertion.

MATRIX MATCHING

41. Correct Answer :

a) Charge of electron → **R) -1.6×10^{-19} C**
Explanation: The magnitude of charge on an electron is -1.6×10^{-19} C.

b) SI unit of electric potential → **T) Volt**
Explanation: SI unit of electric potential is volt (V).

c) SI unit of electric current → **P) Ampere**
Explanation: SI unit of electric current is ampere (A).

d) SI unit of charge → **Q) Coulomb**
Explanation: SI unit of electric charge is coulomb (C).

42. Correct Answer : MATRIX MATCHING

a) When resistors connected in series → **S) Current is constant**
Explanation: In series connection, current remains the same through all resistors.

b) Example for safety devices → **R) MCB and fuse**
Explanation: Fuse and MCB are safety devices used to protect circuits.

c) Household power measures → **electric meter**
Explanation: In household wiring (parallel connection), voltage remains constant across appliances.

d) When resistors connected in parallel → **Q) Voltage is constant**
Explanation: In parallel connection, voltage across each resistor is the same.

COMPREHENSION TYPE

43. Correct Answer : C) Electric power

Explanation: Rate of electrical energy used per unit time is called electric power.
Formula: $P = \frac{E}{t}$.

44. Correct Answer : B) 500 W

Explanation:
Given: $E = 10{,}000\text{J}$, $t = 20\text{s}$

$P = \frac{E}{t} = \frac{10{,}000}{20} = 500\text{W}$

45. Correct Answer : D) Rs. 155

Explanation:
Power = 100 W = 0.1 kW
Daily energy = 0.1 × 10 = 1 kWh
For 31 days (December): 1 × 31 = 31 units
Cost = 31 × 5 = Rs. 155

46. Correct Answer : B) Heat energy

Explanation: Resistance restricts the flow of charges and produces heat due to collision of electrons. Higher resistance produces more heat.

47. Correct Answer : Correct Answer : C) $R \propto \frac{\ell}{A}$

Explanation: Resistance is directly proportional to length (ℓ) and inversely proportional to area of cross-section (A).
Relation: $R \propto \frac{\ell}{A}$.

48. Correct Answer : C) Heating effect of electric current

Explanation: An electric immersion heater works on the heating effect of electric current. Electrical energy converts into heat energy.

INTEGER TYPE

49. Correct Answer : 30 units

Explanation: Total power = 3 × 100 W = 300 W = 0.3 kW
Daily energy = 0.3 × 4 = 1.2 kWh
For 25 days = 1.2 × 25 = 30 kWh
Since 1 kWh = 1 unit

Answer = 30 units

50. Correct Answer : 30 units

Explanation:
Power = 2 kW
Time per day = 45 min = 0.75 hr

Daily energy = 2 × 0.75 = 1.5 kWh
For 20 days = 1.5 × 20 = 30 kWh

Answer = 30 units

51. Correct Answer : 100 W

Explanation:
Energy (E) = 12,000 J
Time = 2 min = 120 s

Power $P = \frac{E}{t} = \frac{12000}{120} = 100\text{W}$

PREVIOUS YEAR QUESTIONS

1. Correct Answer : Circuit with Bulbs L_1, L_2, L_3
Correct Answer: B (F, F, F, T)

Explanation:
False: Bulb L_2 is in the main line, so it gets the most current and is the brightest (not L_1).
False: Bulb L_3 gets half the current of L_2, so it is dimmer than L_2.

False: Current splits between $L_1$and L_3, so they have less current than L_2.
True: Since L_2has twice the current of L_3, it also has twice the voltage (potential difference).

2. Correct Answer : Two Diagrams I and II
Correct Answer: D [(i), (iii) and (iv) only]

Explanation:
Diagram I: Bulbs share the voltage (50V each), so they are dimmer.
Diagram II: Each bulb gets full voltage (100V), so they are brighter.

Since the question asks for INCORRECT statements:
(i) is incorrect (II is actually brighter).
(iii) is incorrect (I will light up, just dimly).
(iv) is incorrect (brightness is not equal).

3. Correct Answer : Electrical Appliance Earthing
Correct Answer: C

Explanation: The earth wire is a safety feature. It provides a path for dangerous current to flow into the ground if there is a loose wire, protecting you from electric shock.

4. Correct Answer : Electromagnet Strength
Correct Answer: A

Explanation: An electromagnet is not permanent. You can change its strength by changing the amount of current or the number of turns in the coil.

5. Correct Answer : Three Identical Bulbs 1, 2, 3 Correct Answer: A (Dimmer, Dimmer, Brighter)

Explanation:
Adding a bulb in parallel makes it easier for current to flow in that section, so Total Current increases.
Bulb 3 is in the main line, so it gets all that extra current and becomes Brighter.
Because Bulb 3 uses up more "energy" (voltage), there is less left over for Bulbs 1 and 2, so they get Dimmer.

6. Correct Answer : Switch Diagram
Correct Answer: A (P, R and T only)

Explanation: To light only Bulb Y:
Switch P (Main power) must be ON.
Switch R (Branch for Y) must be ON.
Switch T (Return path) must be ON.
Switches Q and S must be OFF to keep the other bulbs dark.

7. Correct Answer : Household Circuit Error
Correct Answer: C

Explanation: For safety, the Switch must always be connected to the Live wire. In the diagram, it is on the Neutral wire, which is dangerous because the appliance would still be "live" even when turned off.

8. Correct Answer : Magnetic Field of Wire
Correct Answer: B

Explanation: A magnetic field around a wire is created by the flow of electricity. If you turn off the current, the magnetic field disappears immediately.

9. Correct Answer : Four Bulbs P, Q, R, S (Question 15)
Correct Answer: C

Explanation:
Bulbs P and Q are on a separate branch from Bulb R.
If Bulb Q burns out, that top branch breaks, but the path through Bulb R is still connected, so R stays lighted.

(Note: Options A, B, and D describe bulbs going off that are on independent functioning paths, so C is the only logical choice regarding connectivity).

10. Correct Answer : Matching:

(a) → (iii) Mica
(b) → (iv) Coil
(c) → (i) CFL
(d) → (ii) Nichrome

6. Earth, Moon, and the Sun

Solutions

WORKSHEET – 1

1. **Correct Answer : B) 24 hours**

   Explanation: Earth takes about 24 hours to complete one full rotation on its axis. This rotation causes the regular cycle of day and night on Earth.

2. **Correct Answer: C) Rotation — 24 hours**

   Explanation: Rotation is Earth's spinning motion and it takes about 24 hours. Revolution is different because it takes about 365 days around the Sun.

3. **Correct Answer: A) East to west**

   Explanation: Earth rotates from west to east, so celestial objects appear to move in the opposite direction. This is why the Sun and stars seem to travel from east to west.

4. **Correct Answer: B) Rotation**

   Explanation: Daily changes in the sky occur because Earth rotates continuously. As Earth spins, different parts face the Sun, causing apparent movement of celestial objects.

5. **Correct Answer: B) Movement of Earth around the Sun**

   Explanation: Revolution refers to Earth moving around the Sun along its orbit. This motion takes about one year and is responsible for seasons.

6. **Correct Answer: C) North Pole and South Pole**

   Explanation: Earth rotates around an imaginary line called the axis. This axis passes through the North Pole and South Pole.

7. **Correct Answer: A) Day and night would not occur normally**

   Explanation: Day and night happen because Earth rotates. If rotation stopped, one side of Earth would face the Sun continuously while the other stayed dark.

8. **Correct Answer: B) Earth's rotation**

   Explanation: The stars appear to move across the sky because Earth is rotating. The stars are not actually moving around Earth each night.

9. **Correct Answer: A) East to west**

   Explanation: Due to Earth's west-to-east rotation, celestial objects appear to move from east to west when observed from the Equator.

10. **Correct Answer: B) Orbit**

    Explanation: The orbit is the path Earth follows while revolving around the Sun. It is slightly elliptical and defines one year.

WORKSHEET-2

1. **Correct Answer: B) Earth's tilted axis and spherical shape**

   Seasons occur because Earth's axis is tilted while revolving around the Sun. This tilt causes different parts of Earth to receive different amounts of sunlight during the year.

2. **Correct Answer: C) Sunlight falls more directly**

   and for longer duration
   In June, the Northern Hemisphere is tilted toward the Sun. Therefore, sunlight is more direct and the duration of daylight is longer, making it warmer.

3. **Correct Answer: B) The hemispheres receive** different sunlight angles due to tilt
Because Earth is tilted, one hemisphere faces the Sun while the other tilts away. This difference in sunlight angle causes opposite seasons in the two hemispheres.

4. **Correct Answer: B) 21 June**
Around 21 June (summer solstice), the Northern Hemisphere is tilted most toward the Sun. This results in the longest day and shortest night of the year.

5. **Correct Answer: C) Tilted toward the Sun**
When a hemisphere is tilted toward the Sun, it receives sunlight for more than 12 hours. This leads to longer days and generally warmer conditions.

6. **Correct Answer: B) Shadows form due to orbital positions**
Eclipses occur when the Sun, Earth, and Moon align in a straight line. One object blocks sunlight and casts a shadow on another.

7. **Correct Answer: B) Looking directly at the Sun** can damage eyes
Even during an eclipse, harmful solar rays can damage the retina. Proper eye protection is necessary to safely observe solar eclipses.

8. **Correct Answer: C) Just before or just after totality**
The Diamond Ring Effect appears when a small bright part of the Sun is still visible as totality begins or ends, creating a shining ring-like effect.

9. **Correct Answer: B) 21 March and 23 September**
These dates are equinoxes when the Sun shines equally on both hemispheres. As a result, day and night are nearly equal in length worldwide.

10. **Correct Answer: C) It gradually decreases**
After 21 June, the Northern Hemisphere slowly tilts away from the Sun. This causes daylight hours to decrease until 22 December.

WORKSHEET-3

1. **Correct Answer: D) Both Mercury and Venus**
Mercury and Venus are the only inner planets that do not have any natural satellites. Earth has one moon and Mars has two moons, so both Mercury and Venus have none.

2. **Correct Answer: C) They have small metallic cores and thick atmospheres**
Outer planets are called gas giants because they are very large and mostly made of gases. They have small solid cores surrounded by thick gaseous layers.

3. **Correct Answer: C) Ganymede – Jupiter**
Ganymede is the largest moon in the solar system and it orbits Jupiter. The other pairs are incorrect because those moons belong to different planets.

4. **Correct Answer: C) Methane in its atmosphere**
Methane gas absorbs red light and reflects blue-green light. This is why Uranus appears blue-green when viewed from space.

5. **Correct Answer: C) 2**
Mars has two small moons named Phobos and Deimos. They are irregular in shape and much smaller than Earth's moon.

6. **Correct Answer: B) Venus**
Venus is the hottest planet due to its thick carbon dioxide atmosphere. This atmosphere traps heat strongly through the greenhouse effect.

7. **Correct Answer: C) Venus**
Venus rotates very slowly, taking more time for one rotation than for one revolution around the Sun. Therefore, its day is longer than its year.

8. **Correct Answer: C) 149.6 million km**

Earth's average distance from the Sun is about 149.6 million km. This distance is called one Astronomical Unit (AU).

9. **Correct Answer: B) Fast spinning speed**

Outer planets rotate very quickly on their axes. Because of this fast rotation, they have shorter days compared to many inner planets.

10. **Correct Answer: D) Neptune**

Neptune has the strongest winds in the solar system, reaching extremely high speeds. These powerful winds occur due to its atmospheric conditions and internal heat.

SINGLE OPTION CORRECT TYPE

1. **Correct Answer: B) 59**

Mercury takes about 59 Earth days to rotate once on its axis. Since Earth rotates once every day, Earth completes 59 rotations during that time.

2. **Correct Answer: C) The apparent sizes of Sun and Moon are nearly equal**

The Sun is much larger but also much farther away than the Moon. Because of this, both appear nearly the same size in the sky, allowing eclipses.

3. **Correct Answer: C) 90°**

Earth rotates 360° in 24 hours, so it rotates 15° per hour. In 6 hours, it rotates 15 × 6 = 90°.

4. **Correct Answer: A) 1 : 1.5**

Mercury's year is about 88 Earth days and its day is about 59 Earth days. The ratio is approximately 1 : 1.5.

5. **Correct Answer: A) The Sun's bright surface overwhelms the corona**

The corona is very faint compared to the bright photosphere. It becomes visible only when the Sun's bright surface is fully blocked during totality.

6. **Correct Answer: C) The Northern Hemisphere is tilted away from the Sun**

Seasons occur due to Earth's tilt, not distance from the Sun. In January, less direct sunlight reaches the Northern Hemisphere, causing winter.

7. **Correct Answer: B) Seasonal variations would almost entirely disappear**

Earth's tilt creates seasonal changes. Without tilt, sunlight distribution would remain nearly constant, so seasonal differences would be minimal.

8. **Correct Answer: A) Sunlight bends through the Earth's atmosphere**

Earth's atmosphere bends sunlight and scatters shorter wavelengths. Red light reaches the Moon, giving it a reddish appearance during a lunar eclipse.

9. **Correct Answer: B) Earth**

Earth has suitable temperature, atmosphere, and liquid water that support life. No other planet is confirmed to have life.

10. **Correct Answer: The correct answer is: C) Saturn**

11. **Correct Answer: D) Dwarf planet**

Pluto is classified as a dwarf planet because it does not clear other objects from its orbital path.

12. **Correct Answer: B) Larger than**

Outer planets are gas giants and much bigger in size compared to the inner rocky planets.

13. **Correct Answer: B) Gas giant planets**

Jupiter, Saturn, Uranus, and Neptune are mainly made of gases and have thick atmospheres.

14. **Correct Answer: D) Venus**

Venus comes closest to Earth during its orbit and is considered Earth's nearest planetary neighbor.

15. **Correct Answer: C) Neptune**

16. **Correct Answer: A) Jupiter, Saturn, Uranus, Neptune**

This is the correct order of the outer planets moving away from the Sun.

17. **Correct Answer: C) Mercury, Venus, Earth, Mars, Jupiter,**

Saturn, Uranus, Neptune
This is the correct sequence of planets from the Sun outward.

18. **Correct Answer: A**

19. **Correct Answer: C) Jupiter**

Jupiter has faint rings made of dust particles, though they are not as bright or visible as Saturn's rings.

20. **Correct Answer: C)**

Less density of the planet
Saturn's average density is lower than water, so theoretically it would float if placed in a large enough body of water.

21. **Correct Answer: A) Sun**

The Sun produces its own light and heat through nuclear fusion, which makes it a star

ONE OR MORE THAN ONE CORRECT OPTION TYPE

22. **Correct Answer: A, B, C**
Explanation: Solar eclipses occur at new moon and lunar eclipses at full moon due to alignment. Partial eclipses happen when alignment is not perfect. Not all eclipses are unsafe to observe directly.

23. **Correct Answer: A, B, D**
Explanation: Rotation causes the daily apparent movement of the Sun. Revolution changes the visible stars during the year. Seasons occur due to revolution combined with Earth's axial tilt.

24. **Correct Answer: A, B, D**
Explanation: Apparent size depends on actual size and distance. The Moon is closer to Earth, so it can appear large enough to cover the Sun. The Sun is actually much larger than the Moon.

25. **Correct Answer: A, B, D**

Explanation: The Diamond Ring Effect appears when a small bright part of the Sun remains visible just before or after totality. This creates a bright "diamond" and a glowing ring around the Moon.

26. **Correct Answer: A, B, D**

Explanation: The poles experience long periods of daylight or darkness due to Earth's tilt. The North and South Poles have opposite conditions. Day and night are not always equal at the poles.

27. **Correct Answer : A, B, D**

Explanation: Earth's tilted axis and spherical shape change the angle of sunlight received, causing seasons. Variation in Earth–Sun distance is not the main reason for seasons.

28. **Correct Answer : A, B, D**

Explanation: The observer's rotation causes stationary objects to appear moving in the opposite direction. Apparent motion occurs because the observer's position changes, not because objects actually move.

ASSERTION AND REASON TYPE

29. **Correct Answer: (A) Both A and R are true and R is the correct explanation of A.**

If Earth stopped revolving but kept rotating, our night side would face the same direction in space every night, so the same stars would appear at the same time.
The changing star patterns through the year happen because Earth changes position in its orbit around the Sun.

30. **Correct Answer: (D) A is false but R is true.**

In Aryabhata's analogy, Earth moves eastward, not westward; this makes stars appear to move westward.
The reason is correct because stationary objects seem to move opposite to the observer's motion.

31. Correct Answer: (C) A is true but R is false.

Earth's rotation takes about 24 hours, which defines one day.
However, Earth's revolution around the Sun takes about 365 days, not 24 hours, so the reason is incorrect.

32. Correct Answer: (A) Both A and R are true and R is the correct explanation of A.

Both situations involve apparent motion caused by a moving observer rather than actual movement of objects.
Just like trees seem to move from a merry-go-round, celestial bodies appear to move because Earth itself is rotating.

MATRIX MATCHING TYPE

33. Correct Answer:

A–Q, B–P, C–R, D–S

34. Correct Answer:

A–Q, B–P, C–R, D–S

COMPREHENSION TYPE

35. Correct Answer: C) Planetary classification does not guarantee similar physical conditions.

The passage explains that even planets in the same group show major differences (example: Venus, Mercury, Neptune). This shows that classification alone does not mean planets behave similarly.

36. Correct Answer: B) Mercury has a thin atmosphere that cannot retain heat.

Mercury experiences extreme temperatures because its thin atmosphere cannot trap heat. Venus, in contrast, has a thick atmosphere that holds heat, making it hotter overall.

37. Correct Answer: C) Similar location in the solar system does not ensure similar planetary behaviour.

The passage shows that planets near or far from the Sun can still behave very differently. Factors like atmosphere and rotation create different planetary conditions.

38. Correct Answer: B) Atmospheric properties can dominate over distance in determining temperature.
Venus is hotter than Mercury even though it is farther from the Sun, proving atmosphere plays a stronger role than distance alone in controlling temperature.

39. Correct Answer: C) Planet composition and structure influence rotation and satellite systems.

The passage states that gas giants rotate faster and have rings and many moons, showing that composition and structure affect planetary features.

40. Correct Answer: B) It likely shares characteristics with outer planets.

Fast rotation, many moons, and a gaseous composition are typical traits of outer planets. Therefore, the new planet most logically belongs to that category.

INTEGER TYPE

41. Correct Answer:

The winter solstice usually occurs around 21st or 22nd December each year.
On this day, the Northern Hemisphere has the shortest day and longest night because it is tilted away from the Sun.

42. Correct Answer:
The Sun stays visible at the North Pole for about 6 months without setting. This happens because Earth's axis is tilted, causing continuous daylight during the polar summer.

43. Correct Answer:

Saturn has about 145 known moons (number may change as new ones are discovered).
Its strong gravity captures many natural satellites, making it the planet with one of the largest moon systems.

PREVIOUS YEAR QUESTIONS

1. Correct Answer: C — I, III and V only

- I Morning & Evening Star — correct
- III Earth's twin — correct
- V Hottest planet — correct

2. Correct Answer is B)

(i) East, (ii) West, (iii) North, (iv) Southern.

3. Correct Answer is B)

During a full moon, the Earth lies between the Sun and the Moon.

So the order is:

Sun — Earth — Moon

Written using letters: S E M (SEM)

During a new moon, the Moon lies between the Sun and the Earth.

So the order is:

Sun — Moon — Earth

Written using letters: S M E (SME)

7. Climate, Winds and Cyclones

Solutions

WORKSHEET – 1

1. **Correct Answer: B) Day-to-day condition of theatmosphere at a particular place**

   Explanation: Weather describes short-term atmospheric conditions like temperature, rainfall, wind, and humidity at a specific place. It changes frequently.

2. **Correct Answer: C) Weather changes quickly, but climate changes slowly**

   Explanation: Weather varies daily, while climate is the long-term average pattern observed over many years. Climate changes gradually.

3. **Correct Answer: B) To make scientific weather predictions using data**

   Explanation: The meteorological department collects and analyses atmospheric data to forecast weather. It studies weather but does not control it.

4. **Correct Answer: C) Maximum–Minimum Thermometer**

   Explanation: This instrument records the highest and lowest temperatures reached in a day. It helps compare daily temperature variations.

5. **Correct Answer: B) Rain Gauge — rainfall measurement**

   Explanation: A rain gauge is used to measure the amount of rainfall received in an area. It gives rainfall data accurately.

6. **Correct Answer: B) Humidity is higher, reducing sweat evaporation**

   Explanation: High humidity slows the evaporation of sweat from the body, making people feel hotter and uncomfortable.

7. **Correct Answer: B) Location, altitude, distance from sea, and winds**

   Explanation: These geographical factors influence temperature and rainfall patterns, leading to different climatic regions.

8. **Correct Answer: B) Coastal region**

   Explanation: Coastal areas have moist air due to nearby seas, causing high humidity and sweaty conditions in summer.

9. **Correct Answer: B) Temperature differences in surroundings**

   Explanation: Unequal heating of land and water creates pressure differences, which change wind direction during the day.

10. **Correct Answer: B) Water vapour from seas mixes with air**

    Explanation: Evaporation from seas adds moisture to the air, making coastal regions more humid.

WORKSHEET-2

1. **Correct Answer: C) Moving air in the surroundings**

   Wind is air that moves from one place to another due to pressure differences in the atmosphere.

2. **Correct Answer: C) Air trapped inside the glass**

   prevents water from entering.
   The trapped air occupies space inside the glass and stops water from reaching the paper.

3. **Correct Answer: B) Heated air expands, increasing pressure inside the bottle.**

When air is heated, it expands and pushes outward, making the water drop move outward.

4. **Correct Answer: B) air contracts, reducing pressure inside.**

Cooling makes air contract, lowering pressure inside the bottle, so the drop moves inward.

5. **Correct Answer: B) Because hot air becomes less dense after expansion**

Hot air expands and becomes lighter than cold air, so it rises upward.

6. **Correct Answer: B) Decreases**

Fast-moving air creates lower pressure above the card, allowing higher pressure below to push it up.

7. **Correct Answer: B) Pressure difference is very large**

A larger pressure difference causes faster air movement, producing strong winds.

8. **Correct Answer: B) From land to sea**

At night, land cools faster and develops higher pressure, so air flows from land toward the sea.

9. **Correct Answer: B) Formation of winds**

Uneven heating creates pressure differences that cause air to move, forming winds.

10. **Correct Answer: B) Land heats faster than sea and warm air rises above land**

Warm air over land rises and cooler air from the sea moves in to replace it, creating sea breeze.

WORKSHEET-3

1. **Correct Answer: B**

Fast-moving air creates low pressure above the straw. Higher air pressure on the water pushes it upward into the straw.

2. **Correct Answer: B**

Warm air rises and creates a low-pressure area. Surrounding air moves toward this low-pressure region.

3. **Correct Answer: C**

As air rises, it cools and water vapour condenses. This condensation forms clouds.

4. **Correct Answer: C**

Air moves from high-pressure areas to low-pressure areas. This movement produces strong winds during cyclones.

5. **Correct Answer: B**

Meteorological departments provide accurate weather forecasts and warnings. Following them helps people stay safe.

6. **Correct Answer: B**

Storms can damage electrical systems and cause short circuits. Switching off mains prevents electrical accidents.

7. **Correct Answer: B**

Satellites capture images of cloud systems from space. These images help track cyclone formation and movement.

8. **Correct Answer: B**

Air rushes from high to low pressure, creating strong winds. Earth's rotation makes these winds spin into a cyclone.

9. **Correct Answer: C**

An anemometer measures wind speed using rotating cups or blades. Faster rotation indicates stronger wind.

10. **Correct Answer: B**

Weather data is first collected and analyzed. After analysis, warnings are issued to the public.

SINGLE ANSWER CORRECT TYPE:

1. **Correct Answer: C) A sunny morning suddenly turns rainy in the afternoon**

Weather refers to short-term atmospheric changes. Sudden changes within a day show weather, not climate.

2. **Correct Answer: B) Maximum–Minimum thermometer –Temperature extremes**

This instrument records the highest and lowest temperatures of a day. Other options mismatch the instruments and quantities.

3. **Correct Answer: A) 70 mm**

   1 cm = 10 mm, so 7 cm = 7 × 10 = 70 mm. Rainfall is commonly converted into millimeters.

4. **Correct Answer: C) Weather – short-term atmospheric condition**

   Weather describes daily or hourly conditions, while climate refers to long-term patterns over many years.

5. **Correct Answer: B) Change in weather**

   Rain for a few days is a temporary condition. Climate depends on long-term averages, not short events.

6. **Correct Answer: B) Location and altitude**

   Climate varies mainly due to latitude, altitude, and distance from the sea. Soil or population has little direct effect.

7. **Correct Answer: B) Air inside cools, contracts, and pressure inside decreases**

   Cooling causes air inside the bottle to contract, reducing pressure. This makes the water drop move inward.

8. **Correct Answer: C) burst**

   If internal air pressure increases beyond the balloon's strength, it cannot hold the force and bursts.

9. **Correct Answer: B) rising of a postcard when air is blown above it**

   Fast-moving air creates lower pressure above the card, so higher pressure below pushes it upward.

10. **Correct Answer: C) climate**

    Long-term hot and dry conditions describe climate. Weather would refer to daily changes.

11. **Correct Answer: A) sweat evaporates slowly**

    High humidity means more moisture in air, reducing evaporation of sweat, which makes us feel uncomfortable.

12. **Correct Answer: A) Radar**

    Radar helps scientists detect cyclones early by tracking clouds, rainfall, wind movement, and storms over large areas. It sends radio waves and receives signals back to monitor weather changes.

13. **Correct Answer: B) 14°C**

    Temperature range = Maximum − Minimum = 38°C − 24°C = 14°C. It shows daily variation in temperature.

14. **Correct Answer: B) 15°C.**

    Temperature range means the difference between the highest and lowest temperatures recorded. Here it is **15°C**.

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

15. **Correct Answer: A, B, D**
    Explanation: Warm ocean water, rising moist air, and continuous moisture supply provide energy for cyclone formation. Cyclones have low pressure at the centre, not high pressure.

16. **Correct Answer: A, B, D**
    Explanation: Air moves from high to low pressure, and greater pressure difference increases wind speed. Rising warm air creates low pressure, helping wind formation.

17. **Correct Answer: A, B, C**
    Explanation: Air pressure makes tyres firm, bursts overfilled balloons, and helps water rise in hand pumps. Seasonal changes are not a direct effect of air pressure.

18. **Correct Answer: A, B, C**
    Explanation: Satellites, radars, and weather stations monitor weather conditions and help predict cyclones. Water bottles have no role in forecasting.

19. **Correct Answer: A, B, C**
    Explanation: Moving air removes water vapour and increases evaporation, so clothes dry faster. Higher humidity slows drying instead of increasing it.

ASSERTION AND REASON TYPE

20. **Correct Answer: C**

    Heated air rises because heating makes air expand and become less dense. The reason is false since heating does **not** make air denser or heavier.

21. **Correct Answer: C**

At night, land cools faster than water, so air moves from land toward sea (land breeze). The reason is false because water cools **slower**, not faster, than land.

22. **Correct Answer: A**

Winds form due to unequal heating of Earth's surface. Different regions receive different amounts of solar energy, which correctly explains wind formation.

23. **Correct Answer: A**

Air-filled tyres stay firm because compressed air exerts pressure in all directions. This uniform pressure supports the tyre walls and maintains shape.

MATRIX MATCHING TYPE

24. **Correct Answer: A – Q, B – P, C – R, D – S**

25. **Correct Answer: A–P, B–Q, C–S, D–R**

COMPREHENSION TYPE

26. **Correct Answer: B) Moving air**

Explanation: Wind is air in motion, as mentioned in the passage. It moves due to differences in temperature.

27. **Correct Answer: A) Temperature differences**

Explanation: Changes in temperature cause air movement, which changes wind direction.

28. **Correct Answer: B) To study weather patterns**

Explanation: Scientists observe wind direction to understand weather conditions and atmospheric patterns.

29. **Correct Answer: B) Air escapes from the bottle**

Explanation: Air occupies space inside the bottle and escapes as bubbles when water enters.

30. **Correct Answer: B) Satellite, radar, anemometer**

Explanation: These tools help scientists study winds and predict cyclones for early warnings.

31. **Correct Answer: C) Low atmospheric pressure**

Explanation: Cyclones form in low-pressure areas where air moves inward and creates strong winds.

PREVIOUS YEAR QUESTIONS

1. **Correct Answer: (C) : Statement (C) is incorrect because,**
increased wind speed is accompanied by decrease in air pressure.

2. **Correct Answer: (B) : During the takeoff of an aeroplane,**
there is a low pressure on the top and a high pressure below. This pressure difference helps in easy take off of an aeroplane.

3. **Correct Answer: (A) : Hot air - Rises up; P → (iv)**
Air pressure - Barometer; Q → (iii)
High wind speed - Reduced air pressure; R → (i)
Wind flow - Uneven heating; S → (ii)

4. **Correct Answer: (C) : As the wind blows over the paper,**

so the pressure above it decreases. This pressure difference makes the paper to fly away.

5. **Correct Answer: (C) : Statement 1 : True, as the**

umbrella held upright.
Statement 2 : False, as high speed wind passing over the umbrella create low pressure.

6. **Correct Answer: (C) : Air moves from a region of**
highpressure to low pressure.

7. **Correct Answer: (B) : Cyclones are accompanied by very**
strong winds and the centre of cyclone is a low pressure area. The diameter of eye at cyclone varies from 10 to 30 km. Also, the centre of cyclone is a cloudless calm area. So, here statements I and III are correct.